

Md Rysul Kabir

Bloomington, IN | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Summary

Computer Science Ph.D. student with experience building and evaluating machine learning systems in generative AI, reinforcement learning, and probabilistic modeling. Works primarily in Python and PyTorch, with experience in LLM post-training, distributed multi-GPU training, policy optimization, and Bayesian inference.

Education

Indiana University Bloomington

Ph.D. in Computer Science (GPA: 3.76/4.0)

Bloomington, IN

Aug 2021 – Present

University of Texas Rio Grande Valley

Master of Science in Computer Science (GPA: 4.0/4.0)

Edinburg, TX

Aug 2019 – Jul 2021

Bangladesh University of Engineering and Technology

Bachelor of Science in Mechanical Engineering (GPA: 3.71/4.0 | Rank: 17/202)

Dhaka, Bangladesh

Feb 2013 – Sep 2017

Technical Skills

Languages: Python, C++, C, SQL, R, MATLAB

ML/Methods: Deep Learning, Reinforcement Learning, LLM Post-Training (SFT, RLHF, RLVR), Bayesian Modeling, NLP

Frameworks: PyTorch, Hugging Face Transformers, NumPyro, JAX, TensorFlow, scikit-learn

Systems/Tools: VERL, FSDP, vLLM, Git, Linux, Slurm/HPC, Docker, GCP, MySQL, PostgreSQL, LaTeX

Research & Professional Experience

Indiana University Bloomington

Research Assistant (AI, Cognitive Computing, & LLMs)

Bloomington, IN

Aug 2021 – Present

- Compared harmful SFT, RLVR, and refusal-feature ablation across open-weight LLMs using behavioral, capability, self-audit, representation, and repair analyses; identified distinct failure modes despite similarly high harmful compliance. [Paper](#)
- Developed a novel deep RL agent with time-scale invariant memory to solve complex temporal decision-making tasks; published at **AAAI 2025**. [Project](#) | [Code](#)
- Implemented hierarchical Bayesian models in NumPyro to infer path integration error parameters and quantify group-level differences between SCD and controls; published in **Science Advances**. [Code](#)

Associate Instructor

Aug 2021 – May 2026

- Served as **Head TA** for *Data Mining* (authored homework and exam questions and coordinated grading) and as a TA for *Machine Learning*, *Applied ML*, and *Applied Algorithms*.

University of Texas Rio Grande Valley

Research Assistant (Reinforcement Learning)

Edinburg, TX

Aug 2019 – Jul 2021

- Proposed “Saltatory Rewards” and integrated generalized advantage estimation (GAE) into A2C, achieving **1.5x faster convergence** in learning complex humanlike motions through deep RL. [Thesis](#)
- Implemented reinforcement learning algorithms (PPO, TRPO, A2C, DQN) from scratch in TensorFlow and benchmarked them against standard baselines. [PGRL](#) | [TDRL](#)

Selected Projects

LLM Post-Training with VERL | Python, PyTorch, FSDP, vLLM, Slurm

- Extended VERL by adding custom rollout and reward logic for multi-GPU SFT/RLVR with FSDP and vLLM.

COVID-19 Literature Mining & Clustering | Python, scikit-learn, NLP

- Clustered 340K+ COVID-19 articles using TF-IDF, dimensionality reduction, K-Means/GMM, LDA, and NMF.

Gaussian Process Regression & Kernel Methods | Python, NumPy, Matplotlib

- Built Gaussian process regression from scratch with RBF/polynomial kernels and log marginal likelihood optimization.

Selected Publications

M. R. Kabir and Z. Tiganj. “Different paths to harmful compliance: Behavioral side effects and mechanistic divergence across LLM jailbreaks.” *arXiv preprint arXiv:2604.18510*, 2026.

M. R. Kabir, J. Mochizuki-Freeman, and Z. Tiganj. “Deep reinforcement learning with time-scale invariant memory.” *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

V. Segen, M. R. Kabir, et al. “Path integration impairments reveal early cognitive changes in subjective cognitive decline.” *Science Advances*, 2025.

J. Mochizuki-Freeman, M. R. Kabir, and Z. Tiganj. “Geometry of abstract learned knowledge in deep RL agents.” *NeurIPS Workshop on Symmetry and Geometry*, 2023.